

Advanced Empirical Finance: Final Exam 2026, Part 2

Minimum-variance investing with high-frequency information

Exam number: 37 & 34

Contributions: Exam number 37: Exercise 1, Exercise 3–4. Exam number 34: Exercise 2, Exercise 5.

Exercise 1: Data handling and the realized-variance estimator

The panel is balanced daily but sparse within the minute, so every stock is placed on a regular 09:30–15:59 one-minute grid with non-traded minutes filled by previous-tick interpolation (last trade carried forward). All variance measures use intraday log-returns (overnight excluded). The universe is the 30 current Dow constituents carried backward to 2008, so the findings describe these survivors rather than a tradeable 2008-vintage strategy.

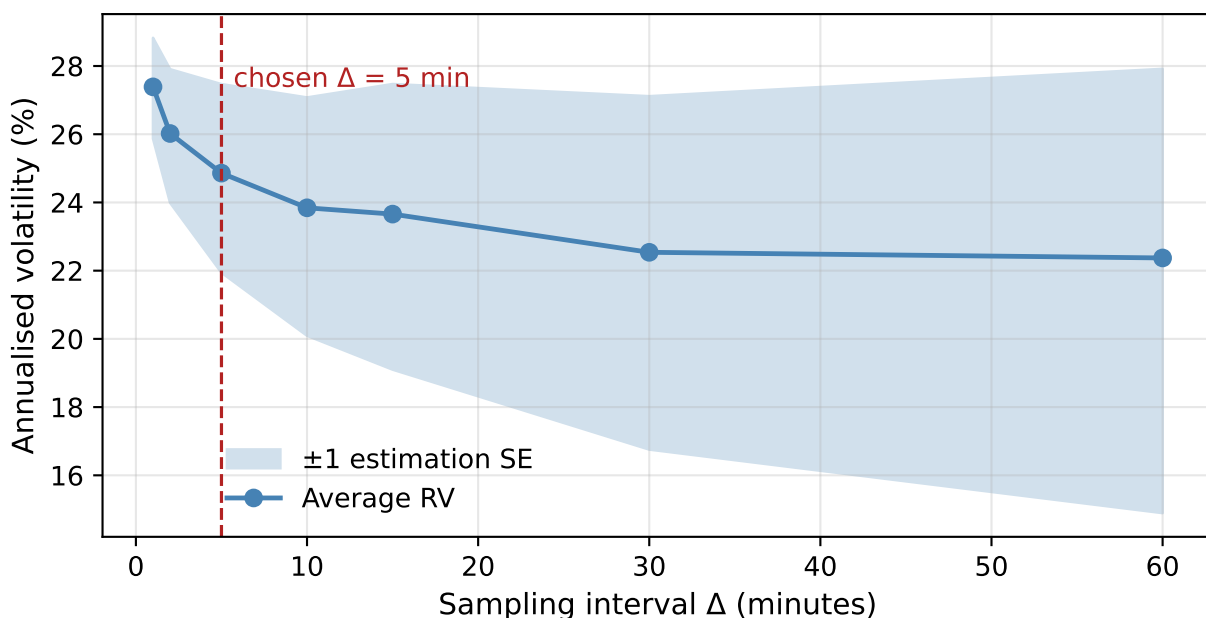


Figure 1: AAPL volatility signature plot: average annualised realized volatility against the sampling interval Δ (line), with the average single-day realized-quarticity standard error (shaded band). The level falls with Δ (microstructure-noise bias) while estimator uncertainty grows (variance).

Figure 1 operationalises the central trade-off from the high-frequency-econometrics lecture. Average realized volatility falls from 27.4% at one minute to roughly 22.4% at 30–60 minutes. Absent frictions, realized variance consistently estimates the integrated variance of the diffusion $dX_t = \mu_t dt + \sigma_t dW_t$ and infill asymptotics favour the highest frequency. Real prices, however, are observed with

microstructure noise, $Y = X + U$, whose sources (bid–ask bounce, price discreteness, asymmetric information, strategic order placement) make U approximately i.i.d. As derived in the lecture, this noise gives returns a negative first-order autocovariance, $\text{Cov}(Y_i, Y_{i-1}) = -a^2$, and biases RV upward by about $2na^2$. Equivalently, the noise share $\pi = 2a^2/(2a^2 + \sigma^2\Delta)$ approaches one as $\Delta \rightarrow 0$, exactly the inflation seen at the finest grid. Coarsening removes this bias but raises the estimator’s variance (fewer increments, the widening band), so an intermediate frequency is optimal. The shaded band quantifies this variance side. Realized variance is asymptotically mixed-normal around the integrated variance, with asymptotic variance governed by the integrated quarticity, which we estimate by the realized quarticity $RQ = (n/3) \sum_i r_i^4$, so that the per-day standard error $\sqrt{2RQ/n}$ is exactly what the band plots. Because the number of increments n falls as Δ grows, this standard error widens monotonically with the sampling interval, the quantitative counterpart of the bias–variance trade-off. Following the in-class discussion and [Liu et al. \(2015\)](#), we adopt the 5-minute realized variance/covariance estimator. By five minutes the curve has flattened (noise negligible) while 78 returns per day retain efficiency. Noise-robust corrections ([Aït-Sahalia et al. \(2005\)](#), or the realized kernel of [Barndorff-Nielsen et al. \(2011\)](#)) are unnecessary here.

Exercise 2: Daily realized covariance

For each day we estimate the 30×30 realized covariance $RC_d = \sum_k r_{d,k} r'_{d,k}$ from synchronised 5-minute returns. Summing outer products makes RC_d positive semi-definite by construction ([Andersen and Bollerslev, 1998](#)). Sampling at five minutes is again the key noise-control choice. It dampens the bid–ask-bounce bias on the diagonal and mitigates non-synchronous-trading bias on the off-diagonal. Figure 2 shows the diagonal series, which exhibits pronounced volatility clustering and near-universal spikes in 2008–09 and March 2020, confirming a strong common volatility factor.

At one minute the off-diagonals would instead be biased towards zero. Stocks do not trade at the same instants, so previous-tick prices are stale and return cross-products misalign (the non-synchronous-trading, or Epps, bias ([Epps, 1979](#))). The lecture tackles this asynchronicity with refresh-time sampling and the multivariate realized kernel of [Barndorff-Nielsen et al. \(2011\)](#), and, for large N , the blocked realized kernel of [Hautsch et al. \(2012\)](#). At $N = 30$, five-minute previous-tick sampling is a simpler synchronisation that already restores economically sensible covariances.

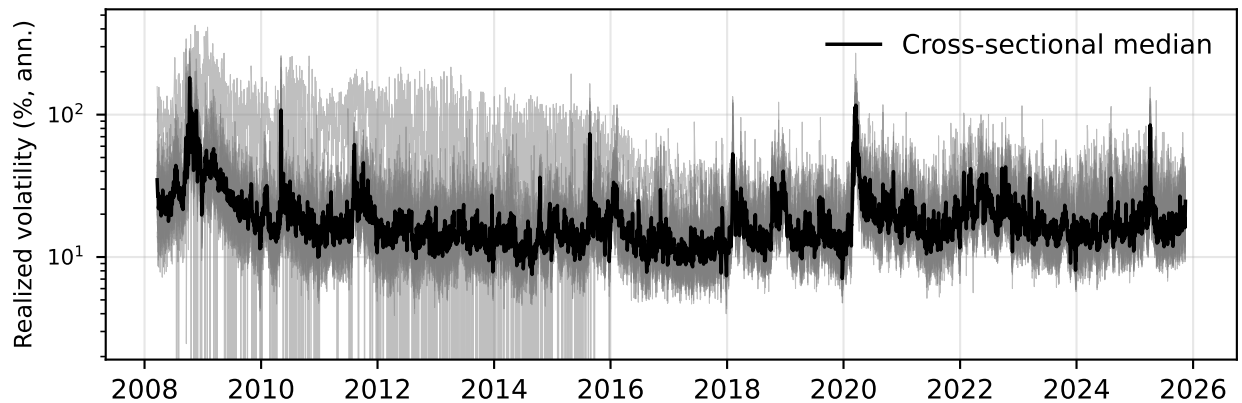


Figure 2: Daily annualised realized volatility (log scale) of all 30 Dow constituents from 5-minute returns. Grey lines: individual stocks; black line: the cross-sectional median. Note the synchronous volatility spikes during the 2008–09 crisis and the March-2020 COVID shock.

Exercises 3–4: Two covariance estimators and minimum-variance weights

Over the second half of the sample (Jan 2017–Nov 2025, 2222 days) we form two one-step-ahead estimators. (a) The high-frequency random walk sets $\hat{\Sigma}_t^{HF} = RC_{t-1}$, imposing martingale (random-walk) dynamics on the covariance matrix, $\mathbb{E}[RC_t | \mathcal{F}_{t-1}] = RC_{t-1}$. This is full persistence with no mean reversion, so the entire previous-day shock is carried into today’s forecast, making it maximally adaptive yet leaving it to inherit a full day of estimation noise. (b) The Ledoit–Wolf estimator shrinks the sample covariance of daily returns over a rolling $W = 252$ -day window towards a scaled-identity target (Ledoit and Wolf, 2004). Setting $W = 252$ (one year) keeps $W > N = 30$ so the matrix stays well-conditioned yet adapts to volatility regimes. Figure 3 shows the LW variances are a smooth, lagged moving average of the jagged realized series, stability traded for responsiveness.

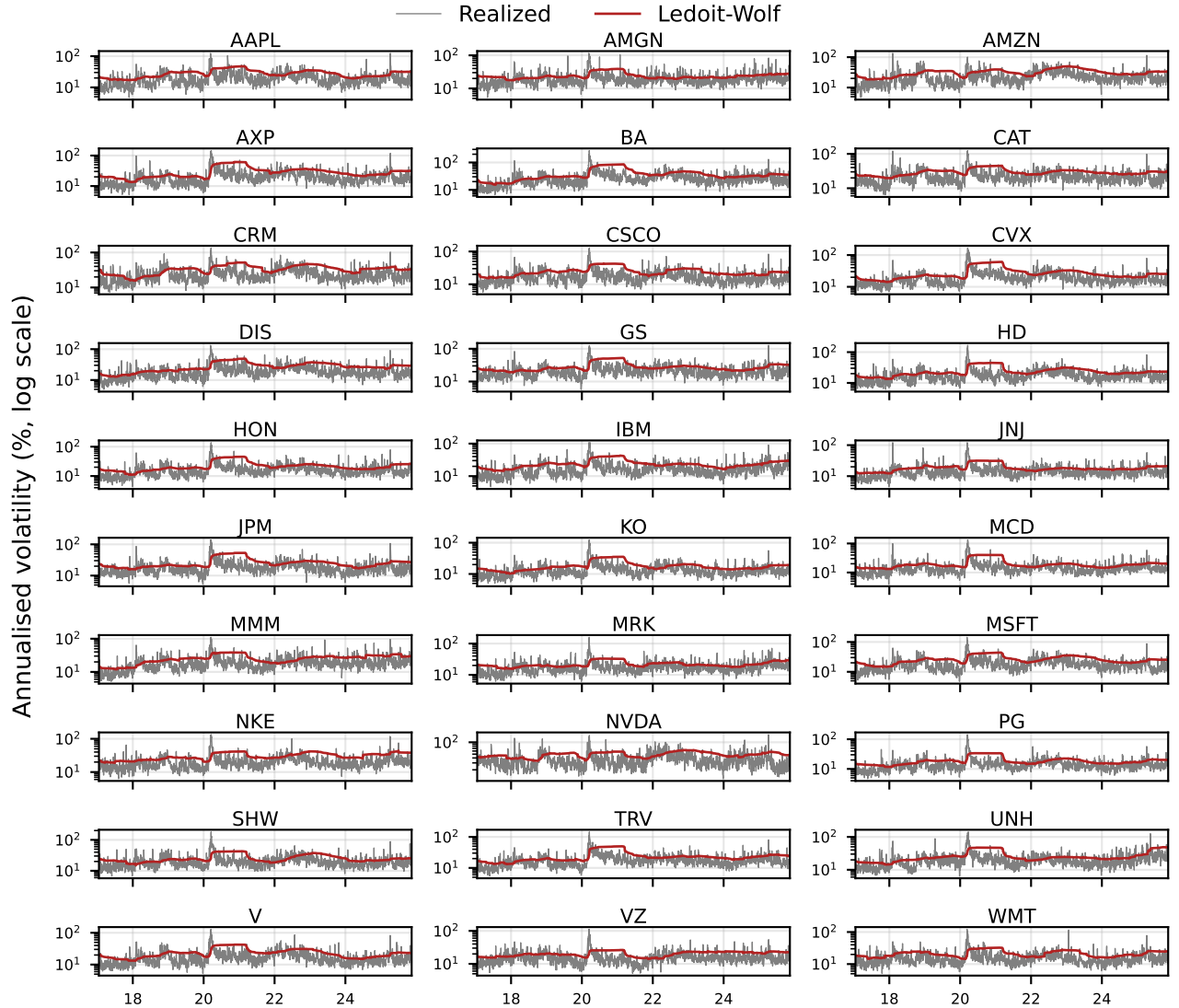


Figure 3: Realized volatility $\sqrt{RC_{t,ii}}$ (grey) versus Ledoit–Wolf volatility $\sqrt{\hat{\Sigma}_{t,ii}^{LW}}$ (red) for each of the 30 stocks over the out-of-sample window (2017–2025; annualised, log scale). The Ledoit–Wolf estimator is a smoothed, lagged moving average of the noisier realized series.

Figure 4 exposes the cost of estimation error in the weights. Inverting a near-singular single-day

matrix, the HF strategy takes extreme, volatile positions (weights -1.17 to 1.48, gross leverage 2.81, turnover 3.49 per day, rebuilding the book daily). LW is far tamer (range -0.24–0.43, turnover 0.071), tilting to low-volatility defensives (VZ, JNJ, MCD) and shorting high-beta cyclicals. The roughly 50-fold turnover gap alone warns that any HF advantage would not survive trading costs.

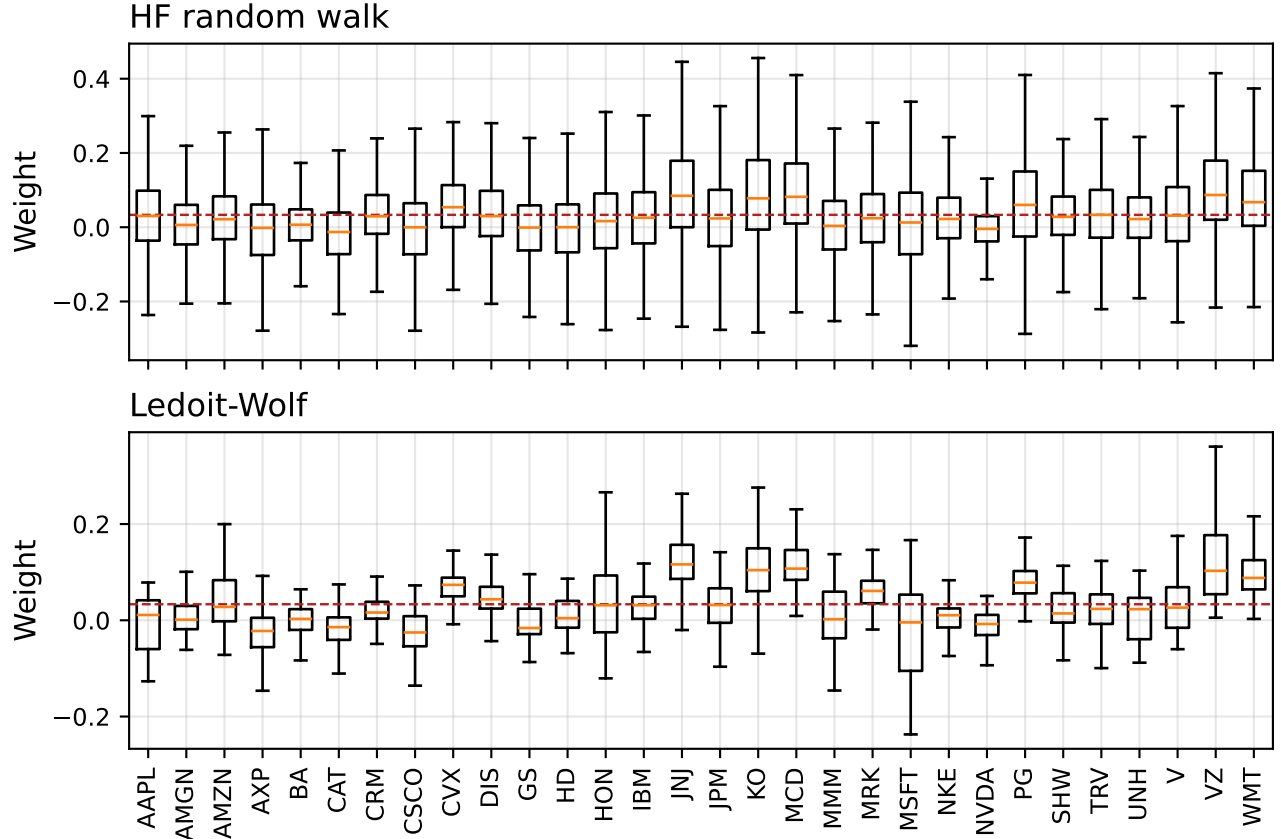


Figure 4: Per-stock distribution of minimum-variance portfolio weights over the out-of-sample window. The HF random-walk strategy (top) takes extreme, unstable long/short positions; the Ledoit-Wolf strategy (bottom) is far more concentrated and stable. Dashed red line: the 1/N weight.

Exercise 5: Out-of-sample evaluation and reflection

This is a genuine out-of-sample test. Every weight at t uses only data through $t - 1$ and is applied to the realised return on t , so Table 1 reports a true walk-forward backtest of the three strategies.

Strategy	Mean excess ret. (% , ann.)	Volatility (% , ann.)	Sharpe	Avg. daily turnover
HF random walk	12.44	16.43	0.76	3.49
Ledoit–Wolf (daily)	11.37	13.99	0.81	0.07
Equal-weight 1/N	17.75	17.87	0.99	0.00

Table 1: Out-of-sample performance of the three strategies, 2017–2025 (risk-free rate assumed zero).

The results are decisive. Both minimum-variance strategies hit their goal, lower realized volatility than 1/N, and Ledoit–Wolf delivers the lowest variance of all (14.0% vs. 16.4% for HF and 17.9% for 1/N). The high-frequency estimator thus gives no measurable advantage. Its single-day, noise-driven matrix produces a higher-variance, higher-turnover portfolio than shrinkage of daily

returns. Realized covariance is an excellent measurement but a poor forecast. Moreover, these are gross returns with no transaction costs charged, so once the roughly 50-fold turnover gap is priced the net comparison widens further against HF. The cost-based case against the high-frequency estimator is therefore conservative. On a risk-adjusted basis the naive $1/N$ portfolio wins the highest Sharpe (0.99), as Figure 5 shows the 2017–2025 bull market rewarded the high-beta names that variance-minimisation under-weights.

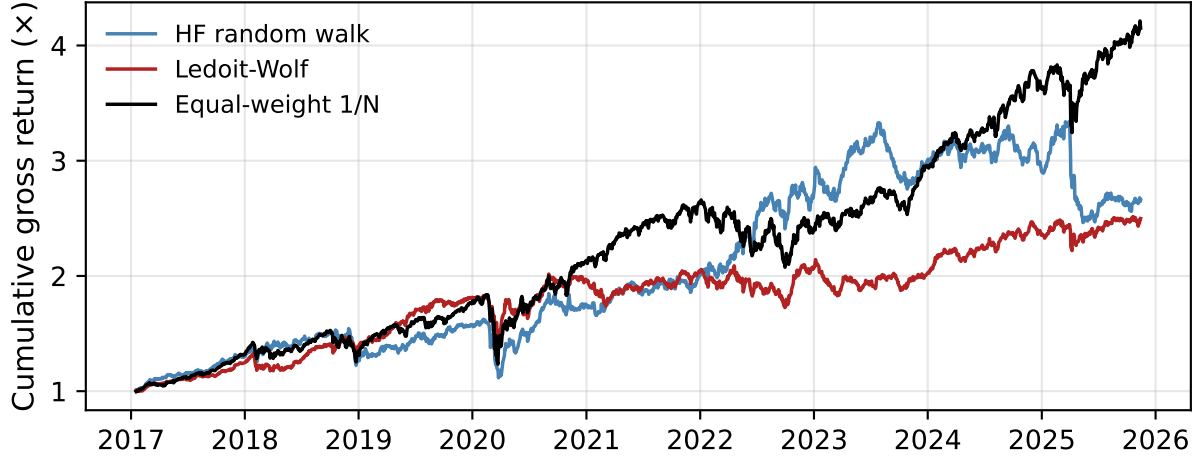


Figure 5: Cumulative gross return of the three strategies over the out-of-sample window (2017–2025): growth of one unit obtained by compounding each day’s realised portfolio return (risk-free rate zero, no transaction costs). The Ledoit-Wolf path is the least volatile, yet equal weighting ends highest in this bull market.

Broader lesson. Extra information (intraday returns, shrinkage targets, factor structures) only pays off when the resulting error reduction outweighs the noise the added flexibility introduces. For $N = 30$ liquid, highly correlated stocks the covariance is already easy to estimate, so the marginal intraday information is small while its noise and turnover are large, and a shrinkage prior dominates. Echoing [DeMiguel et al. \(2009\)](#) and the in-class discussion, statistical sophistication helps mainly when N is large relative to the estimation window, assets are heterogeneous, and transaction costs are honestly charged. Under these conditions, disciplined regularisation, not raw data frequency, does the heavy lifting.

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